Meet 0

October 19, 2026

Algebra

The answer key is below.

p	Answer	p	Answer
1	30	4	13
2	10	5	60
3	40	6	21

1. Alice and Bob are talking to each other. Bob says, "the square root of the number I'm thinking of is a positive integer strictly less than 5." Alice says, "ok rbo" and leaves. If Bob was telling the truth, what is the sum of all possible values of the number he is thinking of?

   Proposed by: Jeff Zhou

   Answer: $\boxed{30}$

   Solution: This is asking for the sum of all positive perfect squares less than $5^2 = 25$, which are 1, 4, 9, and 16. Their sum is

   $$1 + 4 + 9 + 16 = \boxed{30}.$$

2. Two runners start running on a 100 meter track. Runner A runs at a rate of 10 meters per second and runner B runs at a rate of 9 meters per second. Runner A starts 100 meters from the finish, and runner B starts x meters ahead of runner A. If runners A and B reach the end at the same time, compute x.

   Proposed by: Ahmed Ilyasov

   Answer: $\boxed{10}$

   Solution: Note that runner A runs the entire track in 10 seconds, so runner B would need to run the entire track in 10 seconds. Thus, he would need to be 90 meters from the finish line, so $x = 100 - 90 = \boxed{10}$ meters.

3. Steven's clock loses 6 minutes every hour. He sets it to the correct time at exactly noon. Later that day, the clock shows exactly 6:00 PM. How many minutes after 6:00 PM is the actual time?

   Proposed by: Steven Zheng

   Answer: $\boxed{40}$

   Solution: In 60 minutes, Steven's clock runs 54 minutes. Therefore, if Steven's clock shows 6 hours or 360 minutes have passed, in reality, $\frac{60}{54} \cdot 360 = 400$ minutes have passed. Therefore, Steven's clock is $400 - 360 = \boxed{40}$ minutes slow.

4. Given that the sum of the interior angles of an n-gon, $3n$-gon, $5n$-gon, $\ldots$, $89n$-gon equals the sum of the interior angles of a $(2026n - 101)$-gon, find n.

   Proposed by: Taison Scofield

   Answer: $\boxed{13}$

   Solution: We can use the interior angle sum formula. Notice that the 180° cancel out because every term contains it. Therefore, we only care about the $n - 2$ part. This gives

   $$(n - 2) + (3n - 2) + \cdots + (89n - 2) = (2026n - 101) - 2.$$

   The coefficient for n on the left hand side is equal to the sum of the first 45 odd positive integers, which is just $45^2 = 2025$. Hence, we're left with $2025n - 2(45) = 2026n - 103 \implies n = 103 - 90 = \boxed{13}$.

5. Suppose the graphs

$$y = |2x|$$
$$y = -|3x - 90| + c$$

intersect at exactly one point for some value of c. Compute c.

Proposed by: Ahmed Ilyasov

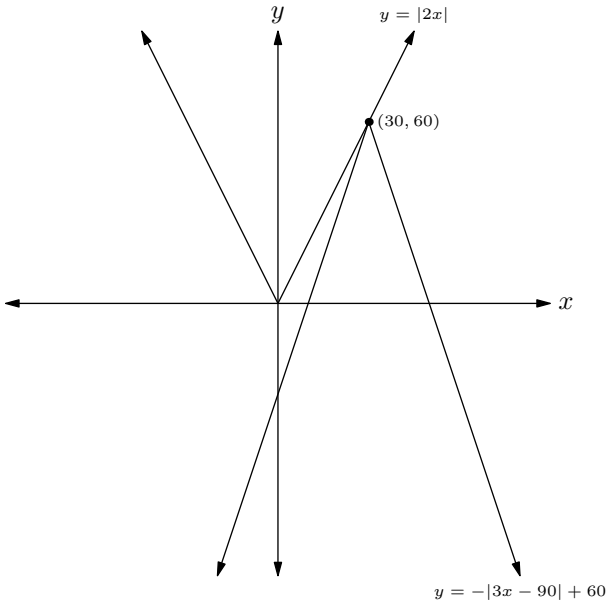
Answer: $\boxed{60}$

Solution: Consider the equation $|2x| + |3x - 90| = c$. We may rewrite the function $f(x) = |2x| + |3x - 90|$ as a piecewise function. Firstly, consider the critical points of this function. For $x \in (-\infty, 0]$, $2x$ is negative and $3x - 90$ is also negative, so $f(x) = -5x + 90$ for those values of x. For $x \in [0, 30]$, $2x$ is positive and $3x - 90$ is negative, so $f(x) = 2x - (3x - 90) = 90 - x$. For $x \in [30, \infty)$, both $2x$ and $3x - 90$ are positive, so $f(x) = 5x - 90$ for those values of x. Thus,

$$f(x) = \begin{cases} -5x + 90, \text{ if } x \leq 0 \\ 90 - x, \text{ if } 0 \leq x \leq 30 \\ 5x - 90, \text{ if } x \geq 30 \end{cases}$$

Now, if $x \leq 0$, $-5x + 90 \geq 90$. If $x \in [0, 30]$, $90 - x \geq 60$. If $x \geq 30$, $5x - 90 \geq 60$. Thus, if $f(x) = c$ has exactly 1 solution, it must occur at the minimum value of $f(x)$, which is equal to $\boxed{60}$.

Alternative Solution: Consider graphing the functions. The function $f(x) = |2x|$ is an absolute value function with vertex at the origin. The function $g(x) = -|3x - 90| + c$ is another absolute value function with vertex at $x = 30$, as the maximum value of $g(x)$ occurs when $-|3x - 90| = 0$, as otherwise, $-|3x - 90| < 0$ and $g(x)$ will decrease. Thus, $g(x)$ has vertex at $(30, c)$. Note that now, c has to lie on the graph of f, as if c lies below f, it never intersects f, and if c lies above f, it intersects f twice. This can be visualized with a graph.



Thus, $|2 \cdot 30| = c \implies c = \boxed{60}$.

6. Let a and b be integers. Suppose that for some positive integer x, $x^{2026} + ax + 1 = 5$ and $x^{2026} + bx + 1 = 17$. Find the sum of all possible integer values of $b - a$.

Proposed by: Ahmed Ilyasov

Answer: $\boxed{21}$

Solution: Solving for a and b, we get that

$$a = \frac{4 - x^{2026}}{x} = \frac{4}{x} - x^{2025}$$
$$b = \frac{16 - x^{2026}}{x} = \frac{16}{x} - x^{2025}.$$

Because a and b are integers, x divides 4 and 16. The positive integers that satisfy this are 1, 2, and 4. Subtracting the equations, we get that $(b - a)x = 12$, so $b - a = \frac{12}{x}$. For the possible values of x, this means that $b - a$ equals 12, 6, and 3. Thus the answer is

$$12 + 6 + 3 = \boxed{21}.$$